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Set 2: Customer Feedback Analysis

Dataset: Customer Satisfaction

Customer ID Age Satisfaction Score

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to display the overall distribution of customer satisfaction scores.
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.
4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

SOLUTION :

CODE :

```
import pandas as pd

import matplotlib.pyplot as plt

from wordcloud import WordCloud

df_feedback = pd.DataFrame({
    'Customer_ID': [1, 2, 3, 4, 5],
    'Age': [25, 30, 35, 28, 40],
    'Satisfaction_Score': [4, 5, 3, 4, 5]
})

plt.figure(figsize=(8, 4))

plt.hist(df_feedback['Age'], bins=5, color='coral', edgecolor='black', alpha=0.8)

plt.title('Distribution of Customer Ages', fontsize=14)
```

```
plt.xlabel('Age', fontsize=12)
plt.ylabel('Frequency', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

```
score_counts = df_feedback['Satisfaction_Score'].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(score_counts, labels=[f'Score {s}' for s in score_counts.index], autopct='%1.1f%%',
        colors=['#66b3ff', '#99ff99', '#ffcc99'], startangle=140)
plt.title('Overall Distribution of Customer Satisfaction Scores', fontsize=14)
plt.tight_layout()
plt.show()
```

```
df_feedback['Age_Group'] = pd.cut(df_feedback['Age'], bins=[20, 30, 40], labels=['20-30',
'31-40'])
ct = pd.crosstab(df_feedback['Age_Group'], df_feedback['Satisfaction_Score'])
```

```
ct.plot(kind='bar', stacked=True, figsize=(8, 4), colormap='viridis')
plt.title('Customer Satisfaction Scores by Age Group', fontsize=14)
plt.xlabel('Age Group', fontsize=12)
plt.ylabel('Customer Count', fontsize=12)
plt.legend(title='Satisfaction Score')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

```
feedback_text = "great service fast delivery user friendly software excellent support highly
satisfied bad UI slow response amazing quality helpful team top notch user experience great
app"
```

```
wordcloud = WordCloud(width=800, height=400, background_color='white',  
colormap='plasma').generate(feedback_text)
```

```
plt.figure(figsize=(10, 5))
```

```
plt.imshow(wordcloud, interpolation='bilinear')
```

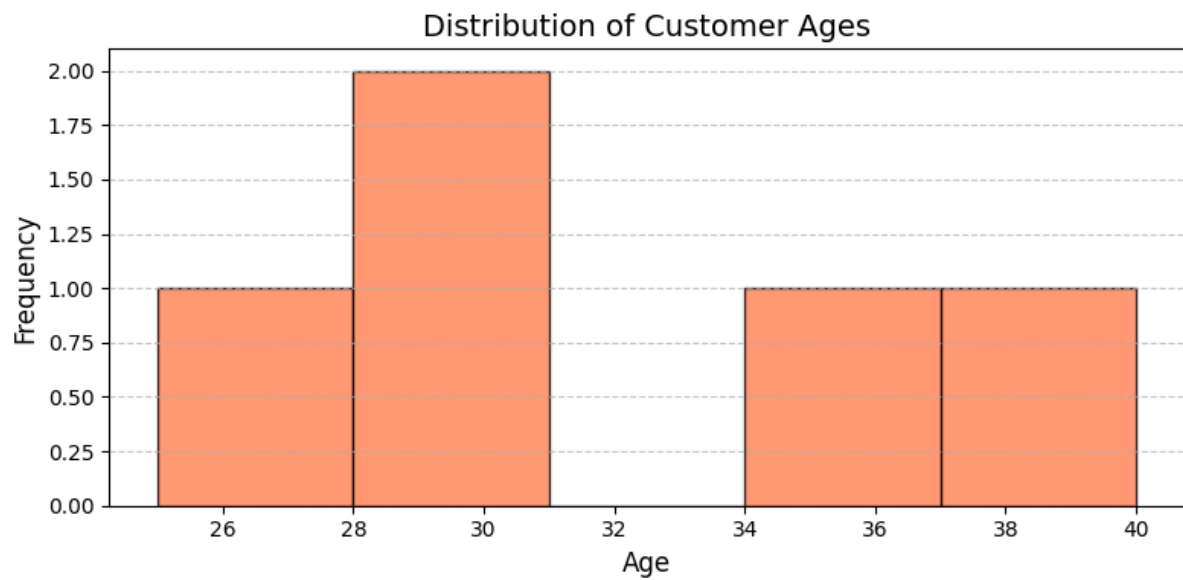
```
plt.axis('off')
```

```
plt.title('Prevalent Customer Sentiments (Word Cloud)', fontsize=14)
```

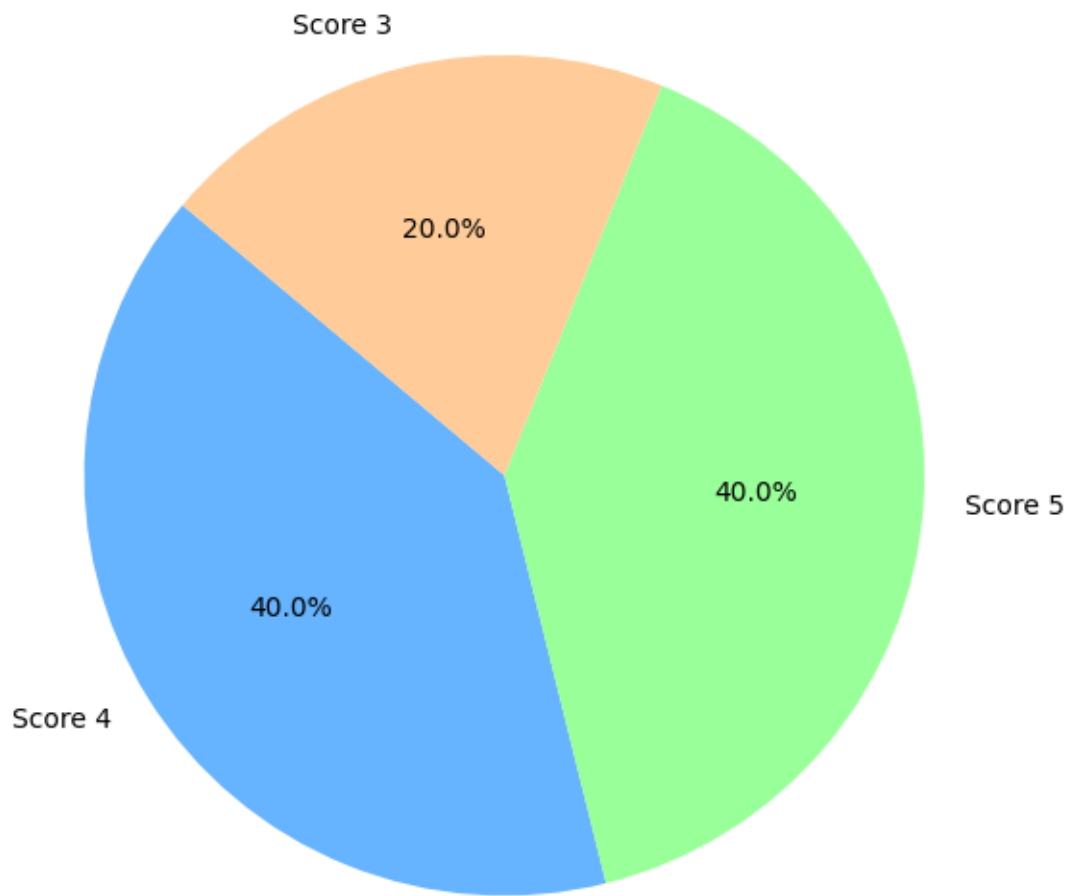
```
plt.tight_layout()
```

```
plt.show()
```

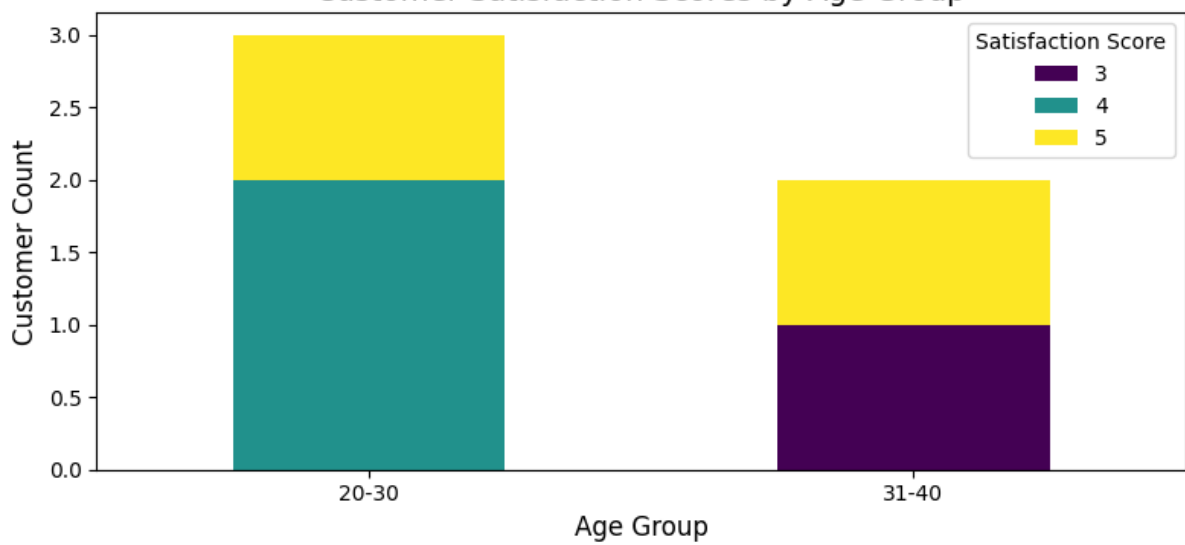
OUTPUT :



Overall Distribution of Customer Satisfaction Scores



Customer Satisfaction Scores by Age Group



Prevalent Customer Sentiments (Word Cloud)

